A Head Camera Is (Almost) All You Need: An Electronics-Free UMI Glove Labelled and Observed by One Moving Camera

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Abstract: Handheld demonstration devices decouple robot data collection from physical hardware, but conventional designs burden the hand with wrist cameras, cables, and complex tracking systems like SLAM or IMUs, while fixed external cameras suffer from severe body occlusion. We show that a single egocentric camera ($122) worn on the collector's head can replace both the wrist camera and the pose tracker. By tracking printed fiducials on an electronics-free glove ($22) relative to paper tiles taped around the workspace, the head camera extracts 6-DoF gripper pose and jaw width. The head naturally tracks the hand without torso occlusion, preserving visual context of both the gripper and task targets. At deployment, the same camera is mounted on a low-cost ($120) robot-adjacent arm that mimics the collector's viewpoint from the robot's joint state. Because collection does not occupy the robot, operators can work in parallel, recording 2.0–3.1 demonstrations per minute compared to 1.6 by teleoperation. Benchmarked against robot encoders, head-camera labels achieve $6.5\,\mathrm{mm}$ median tool-point error and 1.6° rotation error, maintaining 93–$100\,\%$ frame coverage per session (versus 78–$92\,\%$ for an occluded fixed camera). Downstream imitation policies trained on these labels incur only a $3\,\%$ validation loss penalty compared to encoder ground truth. In co-training, adding 194 glove demonstrations to 172 teleoperated episodes reduces ACT validation loss from 0.318 to 0.263; added to 48 teleoperated episodes, it cuts loss from 0.464 to 0.295, matching the performance of 96–172 teleoperated demonstrations alone. On the physical robot, the 48-episode co-trained policy succeeds in 17/25 trials (versus 10/25 for 48 teleoperation episodes alone and 18/25 for the full 172), and the full co-trained head-camera policy succeeds in 20/25, outperforming conventional wrist-camera policies (15/25) that drop objects when task goals leave their narrow view, and fixed-camera baselines (12/25).

1 The idea

Robot teleoperation is fundamentally constrained by hardware availability: collecting each demonstration occupies both the robot and a human operator. Handheld demonstration devices decouple data collection from the robot, allowing operators to gather demonstrations in parallel across arbitrary workspaces. With our system, a collector records demonstrations faster than teleoperation on the same task (2.0–3.1 versus 1.6 demonstrations per minute, §2) without using any robot time. Scaling up collection requires only an inexpensive ($22) printed glove, a wearable ($122) camera, and four paper fiducial tiles taped around the workspace.

Every UMI-style device must resolve two questions: *what does the policy see*, and *how is the gripper tracked*? The conventional approach places both a camera and a pose tracker on the user's wrist.

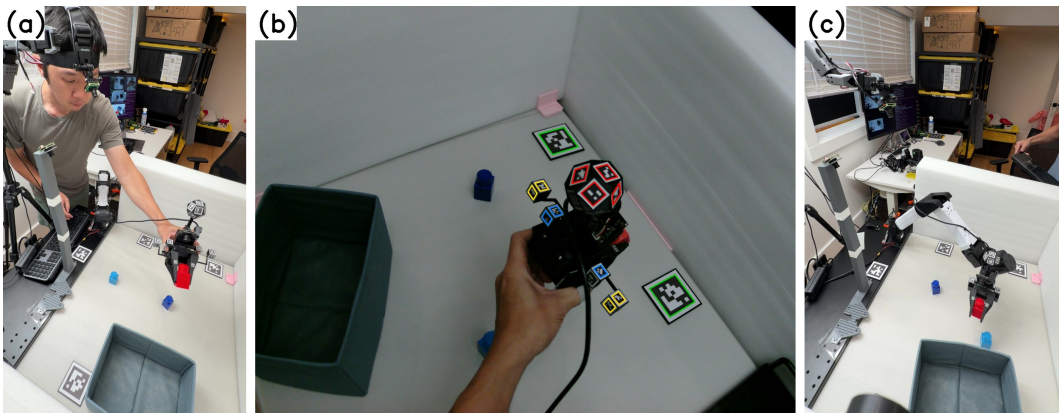


Figure 1: One camera, two mounts. (a) The collector wears the camera on a head strap and carries the electronics-free glove with its fiducial ball over the tiled workspace. (b) The camera's own frame: tiles (green), ball faces (red), jaw tails (blue) and the fixed tags on the gripper back (yellow) give the pose and jaw width. (c) The same camera on the SO-101 positioner beside the YAM during a rollout, same workspace. During collection it is on the collector's head and labels the glove from its own view through the tiles; on the robot the same camera rides a six-joint positioner that follows the gripper. Teleoperated data is recorded through the same positioner, so the three corpora (glove, teleop, rollout) share one viewpoint distribution.

However, existing tracking methods carry substantial drawbacks. SLAM on the wrist camera [1, 2] requires an initial mapping pass and textured scenes, and easily loses track under occlusion or rapid movement. Dedicated visual-inertial sensors add bulk and a second device to the handheld gripper [3]. Mounting a VR controller on the gripper adds significant weight—119 g of YUBI's 319 g total [4, 5]—and relies on external tracking infrastructure with separate calibration errors [6, 7]. Fixed overhead cameras [8] avoid drift, but confine collection to a rigid mount and suffer from severe body occlusion, hiding the hand in 8–22 % of frames (§2). Wrist-mounted observation cameras also present fundamental limitations: they add cables, weight, and synchronisation overhead to the hand. Worse, their narrow field of view focuses strictly on the jaws; during transport, target locations (such as a drop-off bin) easily pass out of frame, sacrificing the global scene context required for manipulation [9].

We address both tracking and observation with a single head-worn camera, directly tackling two central challenges in handheld demonstration [10]: tracking without SLAM and closing the human-to-robot viewpoint gap. Because gaze naturally fixates on objects before manual interaction [11], a collector's head instinctively follows the hand. This natural coordination keeps the gripper centered at arm's length and completely unoccluded by the torso. From this single vantage point, an 11-face fiducial ball on the glove provides the 6-DoF gripper pose, printed jaw tags indicate gripper opening, and four AprilTag tiles [12] taped to the table define the world coordinate frame. As a result, the glove carries zero electronics or cables, while the broad egocentric view keeps both the gripper and the task workspace (including the bin) in frame throughout the demonstration. During deployment, the same camera is mounted on a small, 3D-printed servo arm beside the robot. This positioner imitates human head motion based purely on the robot's joint state, moving identically during teleoperation and autonomous rollouts to ensure that training and deployment share the same viewpoint distribution. In exchange for adding a lightweight camera arm at deployment, we eliminate all onboard wrist cameras, trackers, and cables, while matching teleoperated policy performance.

Two lines of recent work are most relevant. EgoMimic [13] co-trains policies on robot demonstrations and egocentric human video from smart glasses, estimating hand pose directly from video; in contrast, our collector operates a physical replica of the robot gripper, ensuring matched contact physics and directly measured jaw width. Active-vision teleoperation systems [14, 15] steer a robot-mounted camera by tracking an operator wearing a VR headset; in our system, the positioner

operates autonomously during deployment, reproducing human head movements from the robot's joint kinematics without requiring an operator or headset.

2 System

Collection. The glove is a completely unpowered, 3D-printed replica of the YAM gripper [16] from Shin [8], pairing the robot's stock fingers with a rack-and-pinion linear mechanism. To track 6-DoF pose without onboard electronics, the glove mounts an 11-face, $15\,\mathrm{mm}$ ArUco [17] dodecahedron on the back of the hand [18]. Measuring jaw opening from an egocentric head camera requires redesigning the width markers: standard fingertip tags facing a wrist camera [1] remain edge-on or occluded from a head vantage point. Instead, each jaw features a printed wedge—*tail*—displaying $12\,\mathrm{mm}$ AprilTags across five facets alongside a fixed tag on the gripper body. Because at least one tail facet remains visible across hand orientations, the distance between the tail tag and back tag, projected along the jaw travel axis, yields the jaw width on every frame (§2). Omitting the wrist camera, battery, and cabling keeps the glove entirely passive and inexpensive ($22 in parts). The human collector wears an Arducam B0587 4K camera ($122) on a $10 GoPro head strap through a printed GoPro-pattern adapter, capturing footage at $15\,\mathrm{Hz}$ with a short exposure time to prevent motion blur. The camera is aimed once per session: with the live preview open, the collector looks at a point 0.5–$1\,\mathrm{m}$ away and tilts the mount until the centre of the frame coincides with the centre of their own view, so the camera looks where the head looks. Four $60\,\mathrm{mm}$ AprilTag reference tiles define the workspace frame (Fig. 1). These tiles are positioned where the collector's gaze naturally falls during grasping, ensuring at least two remain visible without obstructing the bin or the arm's sweep, while extraneous tags are excluded from the scene. Continuous recording with keypress episode boundaries achieves 2.0–3.1 pick-and-place demonstrations per minute (median episode duration 6–$8\,\mathrm{s}$), outperforming teleoperation on the same arm (1.6 per minute, median $14\,\mathrm{s}$).

Labels from the head camera. Every video frame extracts three geometric signals from a single image: (i) camera pose relative to the workspace tiles; (ii) gripper 6-DoF pose in the camera frame, via fused Perspective-n-Point (PnP) across visible dodecahedron faces; and (iii) jaw opening width, computed from the visible tail tag distance via a unit-calibrated linear map (0.9–$3.8\,\mathrm{mm}$ holdout RMS across calipered spacer holds; calibrated on the robot against motor encoders). Composing the camera and gripper poses gives the tool pose in the workspace coordinate system. A one-time calibration on the robot transforms this into the robot base frame: with the ball mounted on the YAM's gripper, the flange-to-ball link is solved as a robot-world hand-eye problem (OpenCV's `calibrateRobotWorldHandEye`, Shah [19]), and the tile-world-to-base link is a trimmed Kabsch fit of the head-camera tool point (tiles $\times$ ball) to forward kinematics over one 5 065-frame session, enabling joint-space inverse kinematics directly comparable to teleoperated actions. Two mechanisms ensure reliable camera localization despite continuous head motion. First, we refine the nominal tile layout using multi-tile observations extracted directly from demonstration footage. This refitting reduces four-tile reprojection error from $3.4\,\mathrm{px}$ to $0.5\,\mathrm{px}$ RMS, cutting the median disagreement between single-tile camera solves on the same frame from 10–$13\,\mathrm{mm}$ down to 4–$9\,\mathrm{mm}$. Second, frames with only one visible tile are chained to the preceding accepted camera pose under velocity bounds ($0.6\,\mathrm{m/s}$, $120\,^\circ/\mathrm{s}$) rather than discarded. Temporary tracking dropouts longer than $0.3\,\mathrm{s}$ split an episode into segments, and segments under $1\,\mathrm{s}$ are dropped, ensuring momentary occlusions do not invalidate entire demonstrations. Across sessions, this segmentation cuts only 0.7–$7.4\,\%$ of frames, yielding 200 clean segments from 198 demonstrations. In contrast, labelling the same demonstrations with a fixed external camera loses 8–$22\,\%$ of frames due to operator body occlusions, fragmenting the dataset into 259 shorter segments.

The robot's head. To match egocentric viewpoints between demonstration collection and robot deployment, a six-joint SO-101 follower arm [20] stands beside the YAM and carries an identical 4K camera (Fig. 1c). Built from 3D-printed components and six Feetech STS3215 servos ($13.89 each; $121.94 total follower parts list), this active camera positioner adds under $250 to a teleoperation rig. The arm's kinematic model is fitted once from 60 tile-verified poses, achieving $7\,\mathrm{mm}$ / 0.7°

residual error. At the start of each session, the positioner parks at the median head pose observed in glove demonstrations and registers its viewpoint against the reference tiles. During both teleoperation and autonomous rollouts, the positioner tracks the gripper autonomously using only the YAM's joint encoders, remaining completely transparent to the policy. Forward kinematics determines the gripper coordinates; a fixed-gain controller maintains the gripper's initial visual bearing, while a linear residual model fitted to human demonstration sessions reproduces natural head motion relative to hand displacement. Motor commands are low-pass filtered, rate-limited to $50\,^\circ/\mathrm{s}$, and bounded within a safe workspace volume. The resulting camera motion tracks the gripper with $0.4\,\mathrm{s}$ lag and a median tile-referenced position error of $16\,\mathrm{mm}$ ($34\,\mathrm{mm}$ p90), closely replicating human demonstration framing: the normalized gripper centroid in the robot head view (0.37/0.62) matches the human collector's view (0.38/0.60).

Datasets and training. Teleoperation under the head-camera positioner produced 178 episodes (38.6k frames at $15\,\mathrm{Hz}$), with 6 held out as a universal validation set. Four glove sessions yielded 198 demonstrations, resulting in 200 segments after blind cutting (50.0k frames, 6 held out). All policies use Action Chunking with Transformers (ACT) [21] trained through LeRobot [22] for 50k steps with joint-space actions, selecting the best checkpoint by validation loss on the held-out teleoperated episodes. Following Shin [8], glove timelines are resampled to teleoperation pace and measured widths are mapped into the robot's gripper command space. Crucially, anchoring the closed-grasp state to the teleoperators' minimum commanded squeeze (0.22 of full stroke, rather than the 0.25 median) ensures firm grasp transfer: the head-view policies released no block during the carry in 120 rollouts (one slip at lift-off).

3 Experiments

Experimental design. To evaluate our system, we compare eight policy configurations (Table 1) spanning three visual viewpoints: the active head camera, a wrist-mounted fisheye, and a static external camera. In all co-trained conditions, teleoperated demonstrations use robot joint encoders for action labels, while human glove demonstrations are labelled by the head camera (unless specified otherwise for ablation). During teleoperation, the operator viewed the real-time head-camera preview, and all three cameras recorded the demonstrations concurrently; the policies thus share identical demonstration trajectories and differ strictly in observation, with demonstration style shaped by the head-camera view. Table 1 lists the eight arms. All glove-based arms train on the same 198 demonstrations. Arms 1–4 evaluate the scaling benefit of co-training human glove demonstrations across two teleoperation budgets: full (172 episodes) and low (48 episodes). Arm 6 isolates head-camera labelling accuracy by training on teleoperation episodes labelled via the head camera rather than motor encoders. Arms 5, 7, and 8 evaluate camera vantage points. Arm 5 replicates the standard wrist-camera UMI paradigm [1] using a wrist fisheye mounted to the glove for this ablation only, whereas the head-view arms use the glove with no electronics. Arm 7 evaluates a static external camera, training on $10\,\%$ fewer glove frames (43.5k vs. 48.5k) due to human body occlusions. Arm 8 pairs head-camera labels with wrist-camera visual observations, isolating visual perspective from label quality.

Label accuracy. We first evaluate the tracking accuracy of the head camera against robot encoder ground truth by mounting the fiducial ball directly onto the YAM arm. Across 5,065 frames ($91\,\%$ inlier rate), the head-camera pipeline estimates tool-point position with $6.5\,\mathrm{mm}$ median error ($14.1\,\mathrm{mm}$ 90th percentile) and orientation within 1.55° median (2.70° p90), after the calibrated base transform absorbs static mounting offsets. Jaw width estimation was similarly verified across 112 teleoperated episodes (24.6k frames). Visual width tracking achieves a median error of $3.5\,\%$ of stroke ($7.9\,\%$ p90, corresponding to $3\,\mathrm{mm}$ and $7.5\,\mathrm{mm}$ of the $95\,\mathrm{mm}$ stroke), tightening to $2.3\,\%$ median error in the closed state that decides a grasp. The width labels exhibit zero frame lag against motor commands (artificially shifting by one frame strictly raises the error) and achieve full per-frame coverage in 9 out of 10 episodes (lowest: $68\,\%$). Accuracy remains uniform across operating conditions: across bins divided by workspace quadrant, camera distance (0.4–$0.8\,\mathrm{m}$), elevation an-

Table 1: The eight experimental policy variants. Teleoperated actions use robot joint encoders, except Arm 6 which evaluates vision-based labels. Validation loss (ACT loss) is measured on 6 held-out teleoperated episodes with ground-truth encoder targets across all rows. Real-world rollouts: 25 randomized block placements, one trial each (60 s budget); failure modes denote n = no grasp, d = dropped before bin, b = dropped at bin, t = timeout.

#	Arm	Glove labels	Observation	Teleop + glove	Val	Success	Fails
1	teleop only	—	head cam	172	0.318	18/25	n5 d1 t1
2	**device**	head cam	head cam	172 + 194	**0.263**	**20/25**	n5
3	teleop only, low budget	—	head cam	48	0.464	10/25	n15
4	device, low budget	head cam	head cam	48 + 194	0.295	17/25	n8
5	wrist-camera UMI	fixed cam	wrist fisheye	168 + 253	0.257	15/25	n6 d3 b1
6	head-cam teleop labels	—	head cam	172 (head labels)	0.328	15/25	n10
7	fixed camera as system	fixed cam	fixed cam	172 + 255	0.256	12/25	n12 b1
8	head labels, wrist view	head cam	wrist fisheye	168 + 192	0.252	14/25	n3 d7 b1

gle (40–87°), and jaw aspect angle, median error stays between 2.8–4.8 % (6.8–8.5 % p90), rising slightly only when viewing perpendicular to the jaw travel axis. The remaining residual is a systematic 4–6 % under-read at maximum aperture, representing a scale error of the per-session fit (evaluated with alternate-frame holdout) rather than measurement noise; by comparison, the fixed workspace camera in Shin [8] measured the same gripper with 4.4 % median error and 94 % coverage. Downstream policy training confirms that this perceptual accuracy is sufficient for robust imitation learning: training on head-camera visual labels instead of encoders (Arm 6 vs. Arm 1), scored on the same encoder-labelled held-out episodes, incurs only a 3 % increase in validation loss (0.328 vs. 0.318) and a rollout success within the noise of 25 trials (15/25 vs. 18/25; 5 placements won to 8 lost on the same scenes, exact McNemar $p = 0.58$).

Co-training and the teleoperation budget. Following established precedents in co-training across heterogeneous embodiments—such as mixing static and mobile manipulation data [23] or combining human demonstration video with robot data [13]—we train unified policies on merged human glove and robot teleoperation datasets. Adding the 194 human glove demonstrations to all 172 teleoperated demonstrations reduces validation loss on held-out robot episodes from 0.318 to 0.263, a 17 % relative improvement (Arm 2 vs. Arm 1). When teleoperation data is scarce, co-training yields an even larger gain: adding glove demonstrations to 48 teleoperated episodes cuts validation loss from 0.464 to 0.295 (Arm 4 vs. Arm 3). As shown in Fig. 2, log-linear interpolation of the teleoperation-only loss curve demonstrates that the glove dataset provides the equivalent benefit of 65–100 additional teleoperated episodes, achieving an effective exchange rate of 2–3 human demonstrations per teleoperated episode. At the 48-episode budget, co-training saves 50–120 episodes of robot teleoperation, lifting performance to that of 96–172 teleoperation-only episodes. Although the marginal reduction diminishes as the base teleoperation budget grows (a 0.075 loss drop at 24 teleoperated episodes down to 0.011 at 96 episodes, measured against the full mix), glove data effectively stretches a small teleoperated set, though it does not replace the last few dozen teleoperated demonstrations. In physical robot rollouts, the 48-episode co-trained model succeeds in 17/25 trials, outperforming the 48-episode teleoperation baseline (10/25; 9 placements won to 2 lost, $p = 0.065$) and matching the full 172-episode model (18/25; 4 to 5, tied). Every failure of the two 48-episode arms is a missed grasp, and the difference lies in recovery: in Scene 5, the co-trained policy missed its initial grasp, backed away, and successfully re-grasped, whereas the teleoperation-only policy never closed in.

Vantage point comparison. Offline validation loss across the four co-trained camera configurations (Arms 2, 5, 7, and 8) falls within a narrow 0.011 band (0.252–0.263), with each gaining 0.04–0.06 over its teleoperation-only counterpart. Because offline validation loss is known to correlate weakly with physical rollout success [24], real-world rollouts reveal the differences: Arm 2 (head camera) succeeds in 20/25 trials, the wrist-camera baseline (Arm 5) in 15/25, the fixed camera (Arm 7) in 12/25, and head labels on the wrist view (Arm 8) in 14/25. On the shared placements

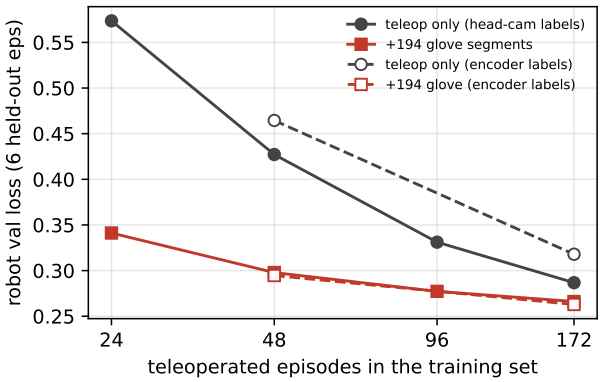


Figure 2: Robot validation loss versus the number of teleoperated demonstrations, with and without 194 co-trained human glove segments. Solid curve: head-camera-labelled teleoperation (intermediate pipeline, four data budgets); dashed curve: encoder-labelled teleoperation (final system, two data budgets). Supplementing teleoperation with glove demonstrations matches the validation loss of 2–4$\times$ as many teleoperated episodes.

the head camera wins 6/1, 11/3 and 8/2 against them ($p = 0.13$, 0.06 and 0.11); no single pair clears 0.05 at 25 placements, but no arm beats Arm 2 on more than three. Failure modes diverge fundamentally by camera vantage point: head-camera policies almost always fail by never grasping, whereas wrist-camera policies frequently fail by dropping the grasped object mid-carry ($d + b = 12$ of 21 failures for Arms 5 and 8, 7 of 11 for Arm 8 alone, compared to 1 of 45 for the head-camera Arms 1–4 and 6, and 1 of 13 for the fixed camera). As illustrated in Fig. 4, this behavior is tied to visual context: in the operator's notes every wrist-view drop occurred while the fisheye was looking away from the receptacle bin during transport, and none while the bin remained visible; the single head-camera drop (Arm 1, Scene 22) occurred immediately after lift-off with both gripper and bin in frame, a slip rather than a release. Because the head camera tracks the hand from an elevated vantage point, it maintains both the gripper and the target bin within a single coherent field of view throughout the carry. Controlling for visual observation (Arm 8 vs. Arm 5) isolates label quality: using head-camera labels rather than external camera labels yields 14/25 against 15/25 (2 placements to 3), so on the wrist view the two label sources are indistinguishable. Conversely, controlling for label source (Arm 8 vs. Arm 2) isolates viewpoint: replacing head-camera observation with wrist-camera observation drops success from 20/25 to 14/25 and converts the no-grasp failures into mid-carry drops (7 of Arm 8's 11 failures). Finally, the static camera (Arm 7) achieved the lowest success of the four co-trained views, 12/25, failing like the head camera by not grasping (12 of 13 failures) but more often; because the active positioner occupied the optimal elevated vantage point, the fixed camera was positioned lower, making its $10\,\%$ occlusion rate an upper bound for static views in this workspace.

Qualitative observations. In Scene 5, Arm 8 (head labels on the wrist view) released the block mid-transport with the block outside its view, as if it no longer registered the object in hand; Arm 5 shows the same pattern in its three mid-carry drops. Head-camera arms that fail do so before contact, by stopping short of the block or closing beside it, and then hold position until the budget expires; they do not release a block once lifted (one slip in 120 trials). Median time to success is 16–19 s for every arm, so the cameras differ in whether the task completes, not in how fast.

4 Limitations and what is next

Our tracking pipeline refines tile positions from a single session using median single-tile poses rather than full bundle adjustment [25]; joint optimization over all multi-tile observations should further reduce single-tile jitter below the current 4–9 mm. Because 6-DoF tracking requires simultaneous

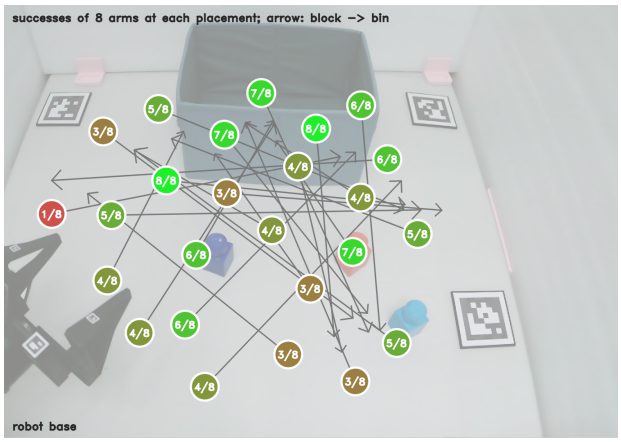


Figure 3: Physical rollout evaluation protocol across 25 randomized workspace placements. For each placement, target blocks are repositioned to identical reference coordinates across policy evaluations using visual ghost alignment (60 s trial budget; execution order randomized per scene). Each disc indicates outcomes across the eight policy arms; the arrow denotes receptacle direction.

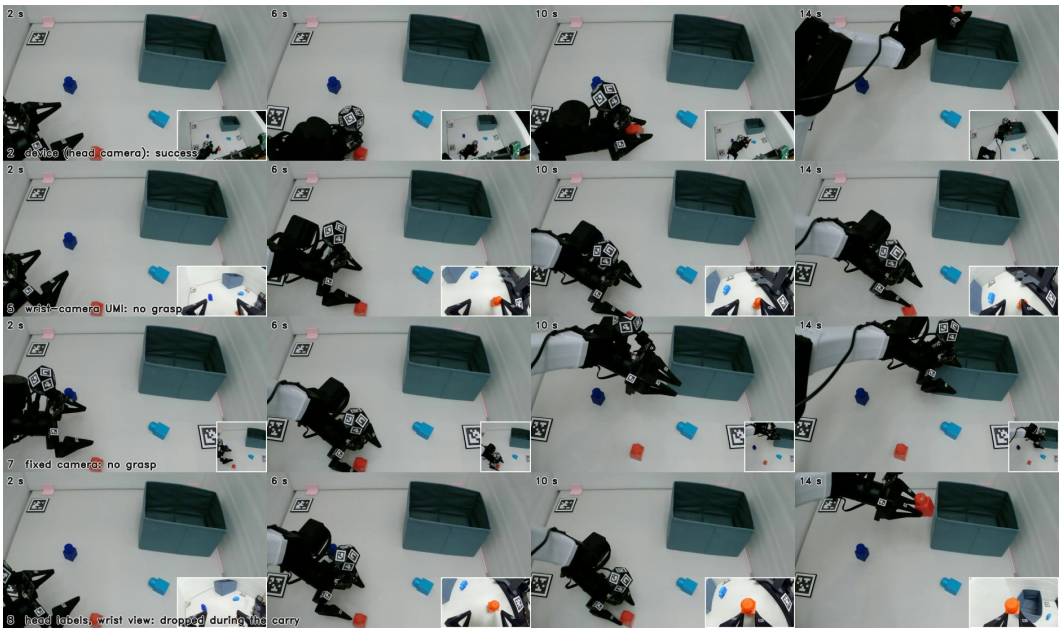


Figure 4: Rollout sequence for Scene 16 across the four camera conditions at $t = 2, 6, 10$, and 14 s, captured from the external workspace camera with policy visual inputs shown inset. During object transport, the wrist fisheye (rows 2 and 4) captures only the grasped block and distant wall, losing sight of the destination bin. In contrast, the head camera (row 1) maintains both the gripper and bin within a single frame, preventing premature release. [Keep or cut for space; fig/make_rollouts.py regenerates for any scene from 15 on.]

visibility of the glove and at least one reference tile, label coverage depends on tile placement; while we establish empirical placement rules (§2), we did not evaluate alternate layouts. On the robot, the camera positioner lags gripper motions by 0.4 s and can reach joint limits during wide sweeping movements. A faster positioner arm or lighter camera payload would reduce this latency and expand the tracking workspace.

An appearance gap also persists between human demonstrations and robot deployment: the head camera captures the collector's forearm and glove fingers, preventing policies trained solely on

glove demonstrations from transferring zero-shot (robot validation loss 0.52 vs. 0.26 co-trained), as observed in the fixed-camera version of the device. Several experimental controls also warrant future evaluation. First, the fixed-camera baseline was mounted low because the positioner occupied the higher vantage point; a higher fixed camera would reduce occlusion, though it still cannot follow the hand. Second, teleoperators guided the robot while viewing the head-camera preview, which may favor egocentric observations; we did not collect a control set teleoperated under a wrist-camera preview. Third, each arm is a single seed evaluated once per placement, so the budget pair and the fixed-camera pair sit at $p = 0.06$ and the remaining pairs above it: the ranking is consistent across every paired comparison, but the individual gaps are not yet significant at 25 placements. Third, although our low-cost design enables parallel collection across multiple operators, all four demonstration sessions were recorded by a single collector with one glove.

Collecting demonstrations in a workspace away from the robot needs no robot-side calibration: the tile-to-base link, fitted here once with the robot holding the ball, can instead be left free and recovered per session, as UMI-style pipelines do with trajectories expressed relative to their start. What the collector must do is lay enough tiles that at least one, preferably two, is in view throughout the task, and tape them down so they do not move within a session. Hardware designs [8] and calibration, recording, and tracking tools [26] are open under the MIT license; robot-side hand-eye calibration and dataset-construction utilities are not yet released. Key next steps include multi-tile workspace mapping, parallel collection across diverse in-the-wild environments, and evaluating the head-camera setup on a second manipulation task.

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